

A New ADS-B Authentication Framework Based on Efficient Hierarchical Identity-Based Signature with Batch Verification

Anjia Yang, Xiao Tan, Joonsang Baek, and Duncan S. Wong

Abstract—Automatic dependent surveillance-broadcast (ADS-B) has become a crucial part of next generation air traffic surveillance technology and will be mandatorily deployed for most of the airspaces worldwide by 2020. Each aircraft equipped with an ADS-B device keeps broadcasting plaintext messages to other aircraft and the ground station controllers once or twice per second. The lack of security measures in ADS-B systems makes it susceptible to different attacks. Among the various security issues, we investigate the integrity and authenticity of ADS-B messages. We propose a new framework for providing ADS-B with authentication based on three-level hierarchical identity-based signature (HIBS) with batch verification. Previous signature-based ADS-B authentication protocols focused on how to generate signatures efficiently, while our schemes can also significantly reduce the verification cost, which is critical to ADS-B systems, since at any time an ADS-B receiver may receive lots of signatures. We design two concrete schemes. The basic scheme supports partial batch verification and the extended scheme provides full batch verification. We give a formal security proof for the extended scheme. Experiment results show that our schemes with batch verification are tremendously more efficient in batch verifying n signatures than verifying n signatures independently. For example, the running time of verifying 100 signatures is 502 and 484 ms for the basic scheme and the extended scheme respectively, while the time is 2500 ms if verifying the signatures independently.

Index Terms—ADS-B, authentication, batch verification, aviation communication security, hierarchical identity-based signature

1 INTRODUCTION

THE number of aircraft has been increasing tremendously over the last decade. According to [23], the average number of registered aircraft movements over Europe is around 26,000 per day. With more and more people choosing to travel by air, we may expect a further boost in air traffic load in the future. At the same time, air traffic control (ATC) and the security of civil aviation encounter large challenges.

Conventional ATC techniques are based on radar systems which include primary surveillance radars (PSR) and secondary surveillance radars (SSR). PSRs are independent and non-cooperative. Namely, PSRs transmit high-frequency signals, receive the echoes reflected from the aircraft and then can determine the position of the aircraft, without requiring the aircraft's participation. On the other hand, SSRs cooperate with and interrogate the aircraft to get responses which are generated by the onboard systems equipped in the aircraft. The responses may contain information of the aircraft such as identification

codes, height, and altitude. However, traditional PSR and SSR systems suffer from some disadvantages such as low precision and high cost.

Recently, a new technique for ATC called automatic dependent surveillance-broadcast (ADS-B) system has been standardized by Federal Aviation Administration (FAA) in America and European Organisation for the Safety of Air Navigation (EUROCONTROL) in Europe. It will replace conventional radar systems and be deployed as part of the next generation air transportation systems. Some countries like Australia and Canada have already deployed ADS-B. The FAA requires aircraft flying within US airspace to be equipped with ADS-B systems by 2020, while the EUROCONTROL mandates ADS-B in European airspace by 2017.

Unlike in traditional radar systems where aircraft only respond to interrogations by ground stations, in the ADS-B system, aircraft continuously obtain their positions based on some satellite positioning techniques (e.g., GPS) and periodically broadcast their positions as well as some other information such as the current velocity to ground stations and other aircraft. With this accurate information, the ground controllers or other surrounding aircraft can monitor and track the location and path of an aircraft, which provides aircraft and the ground controllers a common situational awareness. This improves pilots' decision-making ability dramatically and makes air traffic management much easier.

The ADS-B system has had impact not only on air navigation and communication systems but also web service technology. As discussed in [14], a mashup technique can be employed to link and display real-time ADS-B data from

• A. Yang, X. Tan, and D.S. Wong are with the Department of Computer Science, City University of Hong Kong, Hong Kong, China. E-mail: ayang3-c@my.cityu.edu.hk, xiaotan4@gapps.cityu.edu.hk, duncan@cityu.edu.hk.

• J. Baek is with the Department of Electrical and Computer Engineering, Khalifa University of Science, Technology and Research, Abu Dhabi, UAE. E-mail: joon.baek@kustar.ac.ae.

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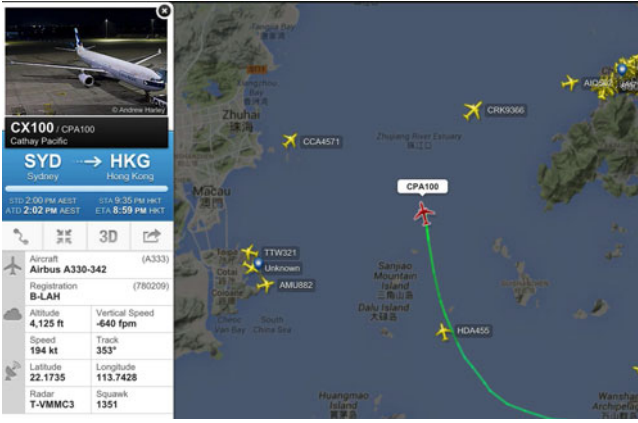


Fig. 1. A mashup displaying ADS-B data of a plane around hong kong, provided by flightradar24.com [1].

planes and various ground entities quipped with ADS-B receivers on the representative web pages. In this respect, aircrafts can be viewed as “smart objects” that can create a graphical representation of themselves on the web. Recently, flight tracker web sites based on the mashup of ADS-B data have gained popularity, providing web users with a visual overview of air-traffic around the world. A snapshot from one of such sites, flightradar24.com [1], is provided in Fig. 1. In this snapshot, a plane object created through mashup of ADS-B data displays the basic ADS-B attributes such as callsign, registration, altitude, speed and position. Providing ADS-B data with authenticity, which is a main theme of this paper, is also important for the related web services to maintain high level of reliability.

Since air traffic management heavily relies on ADS-B messages, vulnerabilities in ADS-B applications should be treated seriously. ADS-B data are broadcast through wireless channel (radio frequency data-link) without any cryptographic mechanisms implemented. Besides the ground controllers and aircraft, anyone who holds a single low-cost ADS-B receiver can obtain the ADS-B data. An active attacker who has full control over the wireless communication channel can inject, modify and delete ADS-B messages [20]. Various kinds of practical attacks on ADS-B devices have been demonstrated by some researchers [7], [15], [16], [17], [20], [21]. Among security issues defined in [21], we mainly deal with data integrity and source integrity. Data integrity ensures that the ADS-B data has not been modified upon arriving at the receivers. Source integrity, or called the authenticity of ADS-B data, makes sure that any ADS-B message originates from a sender that claims to have sent it. Both of these two properties are essential to secure the ADS-B systems, since otherwise an active attacker can modify or inject messages which could result in some destructive attacks such as Ghost Aircraft Injection attack and Virtual Trajectory Modification attack introduced by [20]. In this paper, we propose to apply hierarchical identity-based signature (HIBS) [11] to ADS-B authentication and in fact a three-level HIBS is sufficient. The top level PKG could be an authoritative organization such as the International Civil Aviation Organization (ICAO), the FAA or the EUROCONTROL. The second level consists of different airlines around the world and the aircraft stand in the third level.

In addition, at any time the aircraft or the ground ADS-B receivers will receive a large number of signatures from different surrounding aircrafts which may belong to different airlines and it needs an efficient scheme to verify these signatures as soon as possible. We use batch verification to mitigate this concern. Batch verification introduced by Fiat [10] in 1989, was devised to improve the efficiency of verification process for multiple signatures. According to Yoon et al. [25], batch verification is classified into three types, while we deal with the most intractable but desirable one, i.e., type 3. Type 3 batch verification allows multiple signatures on multiple messages generated by multiple signers to be verified at the same time, where the total number of dominant (e.g., pairing) operations is independent of the number of signatures. The reason we are interested in Type 3 batch verification is that only this type is suitable for the application of ADS-B. Hereafter, whenever we mention batch verification, we refer to the one of Type 3. Depending on whether there are restrictions on those signatures to be verified, batch verification can be classified into full batch verification and partial batch verification, which we will explain later. To the best of our knowledge, there is no HIBS scheme that can achieve batch verification in the literature.

1.1 Contributions

- 1) We propose a new authentication framework for ADS-B systems supporting batch verification to ensure the integrity and authenticity of ADS-B messages.

In this new framework, we employ a three-level HIBS mechanism. The top level PKG generates private keys for the second-level PKGs (the airlines). Each airline is responsible for generating private signing keys for its affiliated aircraft. Each aircraft signs its messages with the corresponding signing key and broadcasts the messages together with the signatures. Upon receiving multiple messages and the corresponding signatures from different aircraft, a verifier (an aircraft or other ADS-B receivers such as the ground station controllers) can verify these signatures at the same time, while the cost of batch verification of n signatures is enormously reduced compared with that of verifying n signatures independently.

- 2) We design two concrete schemes under the framework we defined. The basic scheme can only achieve partial batch verification, which means that multiple signatures coming from different aircraft that belong to the same airline can be verified at one time. However, partial batch verification might be insufficient in practice since the multiple signatures are generally produced by aircraft belonging to different airlines. To address this problem, we extend the basic scheme to achieve full batch verification, that is, multiple signatures coming from different aircraft on different messages can be verified at one time, where the aircraft could belong to different airlines. The disadvantage of the extended scheme is that we need to employ some existing techniques to ensure the integrity of some public values (please refer to the extended scheme in Section 4.2 for the details).

We leave it as an open problem to propose a HIBS scheme that can achieve full batch verification without this requirement.

- 3) We give a formal security proof for the extended scheme in the selective-ID security model with random oracle assumption.
- 4) We implement both the basic and the extended schemes and the results show that our proposed schemes are indeed greatly more efficient in batch verifying n signatures than verifying n signatures independently.

1.2 Related Work

The ADS-B security issues begin to attract people's attention only in recent years. There are two types of countermeasures: non-cryptographic and cryptographic solutions. Non-cryptographic methods include multilateration [13], [21], distance bounding [21], Kalman filtering [21], data fusion [21] and the lightweight location verification proposed by Strohmeier et al. [22].

In terms of cryptographic solutions, some efforts have been made to provide the integrity and authenticity of ADS-B messages in the literature [2], [3], [8], [12], [18], [19]. They either proposed to use symmetric key based authentication methods like MAC or suggested employ asymmetric key based authentication solutions like digital signatures. However, symmetric key cryptography requires the sender and the receiver to pre-share a secret key, which makes its deployment difficult. Digital signatures seem to be a good method but they should satisfy some specific properties in accordance with ADS-B applications. First, the signatures should be short, since the typical payload size of ADS-B message is only several hundred bits. In addition, multiple signatures from different signers should be able to be verified very quickly, since each aircraft keeps broadcasting messages as well as receiving many ADS-B messages from its surrounding aircrafts within each second. RSA signatures satisfy the latter property, but are generally thousands of bits in length which is not suitable for ADS-B application. Recently, the development of pairing-based cryptography makes it possible to produce short signatures which can also be verified in *batch*. Boneh et al. [5] proposed first pairing-based short signature scheme but it is only suitable to verify a bunch of signatures generated by the same singer. Identity-based signature (IBS) schemes like [6] and [9] can batch verify multiple signatures generated by distinct signers. However, it is undesirable to give the burdensome job of generating signing keys for all aircraft to one PKG.

Wesson et al. [24] discussed whether cryptography can secure ADS-B and they concluded that the most practical and effective cryptographic approach is the asymmetric-key elliptic curve digital signature method, and to broadcast ADS-B messages over alternative authentication channels.

Organization. The rest of this paper is organized as follows. Section 2 gives some preliminaries and definitions of ADS-B systems. The security model is defined in Section 3. In Section 4 we propose two schemes and the security proof for the extended scheme is given in Section 5. Section 6 demonstrates the experiment results and finally we make a conclusion of this paper in Section 7.

2 DEFINITIONS

2.1 Mathematical Notations

2.1.1 Bilinear Map

Let $\mathbb{G}$ and $\mathbb{G}_T$ be two cyclic groups of prime order p . Let g be a generator of $\mathbb{G}$. An admissible bilinear map is a map $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ with the following properties:

- 1) Bilinearity: for any $a, b \in \mathbb{Z}_p$, and $u, v \in \mathbb{G}$, $e(u^a, v^b) = e(u, v)^{ab}$.
- 2) Non-degeneracy: If g is a generator of $\mathbb{G}$, then $e(g, g)$ is a generator of $\mathbb{G}_T$, i.e., $e(g, g) \neq 1$.
- 3) Computable: There exists an efficient algorithm to compute $e(u, v)$ for all $u, v \in \mathbb{G}$.

2.2 ADS-B Hierarchical Identity Based Signature

ADS-B HIBS is a three-level hierarchical identity based signature with support of batch verification that targets for the application of ADS-B. It is a tree structure, where the Level-0 node is the root PKG which could be the ICAO, the FAA or the EUROCONTROL. The root PKG keeps the master key pair (msk, mpk) that will be used to generate secret keys for low-level nodes. The Level-1 nodes are the airlines each with an identity like ID_A . The airlines are the second level PKGs which can generate the secret signing keys for the bottom level nodes. The Level-2 nodes are the aircraft each with an identity like ID_F .

An ADS-B HIBS scheme consists of the following algorithms:

Setup: On input of a security parameter κ , the Setup algorithm allows the root PKG to generate the master secret key msk and the master public key mpk .

Extract^A: On input of the master secret key msk , and an airline identity ID_A , the Extract^A algorithm generates a secret key sk_{ID_A} for ID_A .

Extract^F: On input of the secret key sk_{ID_A} for an airline identity ID_A and an aircraft identity ID_F , the Extract^F algorithm generates a secret key sk_{ID_F} for ID_F belonging to ID_A .

Sign: On input of the secret key sk_{ID_F} and a message m , the Sign algorithm generates a signature σ of m .

Verify: On input of a signature σ on a messages m with respect to airline identity ID_A and aircraft identity ID_F , the Verify algorithm outputs either 0 or 1, where 1 represents that the signature is valid.

BVerify: On input of a list of signatures $\sigma_1, \dots, \sigma_n$ on messages $m_1, \dots, m_n$ with respect to airline identities $ID_{A_1}, \dots, ID_{A_n}$ and aircraft identities $ID_{F_1}, \dots, ID_{F_n}$, the BVerify algorithm outputs either 0 or 1, where 1 represents that all of the n signatures are valid. Notice that this algorithm degenerates to the Verify algorithm when $n = 1$.

Definition 1. An ADS-B HIBS scheme with batch verification is said to be correct, if for any $(msk, mpk) \leftarrow \text{Setup}(\kappa)$, the following conditions hold:

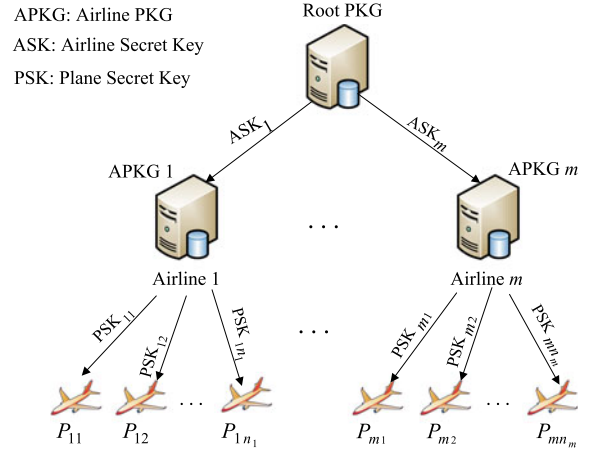
- 1) For any identity ID_A , ID_F and any message m , if $\sigma \leftarrow \text{Sign}(sk_{ID_F}, m)$ where $sk_{ID_F} \leftarrow \text{Extract}^F(sk_{ID_A}, ID_F)$ and $sk_{ID_A} \leftarrow \text{Extract}^A(msk, ID_A)$, then $\text{Verify}(\sigma, ID_A, ID_F, m) = 1$.
- 2) For a list of identities $\{ID_{A_i}\}_{i=1}^n$, $\{ID_{F_i}\}_{i=1}^n$ and messages $\{m_i\}_{i=1}^n$, if $\sigma_i \leftarrow \text{Sign}(sk_{ID_{F_i}}, m_i)$ where $sk_{ID_{F_i}} \leftarrow \text{Extract}^F(sk_{ID_{A_i}}, ID_{F_i})$ and $sk_{ID_{A_i}} \leftarrow \text{Extract}^A(msk, ID_{A_i})$, then $\text{BVerify}(\{\sigma_i\}_{i=1}^n, \{ID_{A_i}\}_{i=1}^n, \{ID_{F_i}\}_{i=1}^n, \{m_i\}_{i=1}^n) = 1$.

$\leftarrow \text{Extract}^F(sk_{ID_{A_i}}, ID_{F_i})$ and $sk_{ID_{A_i}} \leftarrow \text{Extract}^A(msk, ID_{A_i})$, then $\text{BVerify}(\{\sigma_i\}_{i=1}^n, \{ID_{A_i}\}_{i=1}^n, \{ID_{F_i}\}_{i=1}^n, \{m_i\}_{i=1}^n) = 1$.

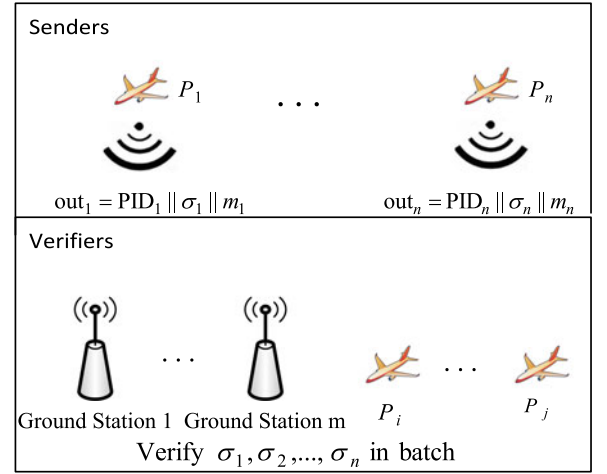
2.3 The Proposed Framework

In this section, we describe our authentication framework for ADS-B systems as follows and graphically shown in Fig. 2.

- 1) The ICAO, the FAA or the EUROCONTROL could act as the root PKG, which runs the Setup algorithm to produce the master secret key msk and the public parameters mpk .
- 2) An airline registers to the root PKG with its identity ID_i . The root PKG invokes the Extract^A algorithm to generate a secret key ASK_i for this airline.
- 3) For each plane P_{ij} belonging to an airline ID_i , the airline ID_i acts as the second-level PKG $APKG_i$ and runs the Extract^F algorithm to generate a secret key PSK_{ij} for this plane. Note that PSK_{ij} may be stored in the plane's electronic system.
- 4) Upon sending out a message m , an aircraft runs the Sign algorithm to produce a signature σ corresponding to this message and broadcasts $out_i = (PID_i, \sigma_i, m_i)$, where PID_i is the plane's unique identity.
- 5) Upon receiving multiple message-signature pairs, the receiver (including the other aircraft, the ground controllers or any other parties who are equipped with an ADS-B receiver) runs the BVerify algorithm to batch verify these signatures. If the output is 1, it means the integrity and authenticity of the received messages are ensured.



(a) Setup and Extract Phase



(b) Sign and Verify Phase

3 SECURITY MODEL

For ADS-B HIBS, the existential unforgeability against selective ID and chosen message attacks can be formalized as the following adversarial game between a challenger $\mathcal{C}$ and a probabilistic polynomial time (PPT) adversary $\mathcal{A}$:

Setup phase: $\mathcal{A}$ sends the target Level-1 identity ID_A^* and Level-2 identity ID_F^* to $\mathcal{C}$. $\mathcal{C}$ runs $\text{Setup}(\kappa)$ to generate the master key pair (msk, mpk) , then sends mpk to $\mathcal{A}$.

Query phase: $\mathcal{A}$ can adaptively make the following three types of queries within polynomial times:

- **Extract^A Query:** When $\mathcal{A}$ queries for the secret key of an airline ID_A , $\mathcal{C}$ returns $sk_{ID_A} \leftarrow \text{Extract}^A(msk, ID_A)$.
- **Extract^F Query:** When $\mathcal{A}$ queries for the secret key of an aircraft ID_F belonging to an airline ID_A , $\mathcal{C}$ returns $sk_{ID_F} \leftarrow \text{Extract}^F(sk_{ID_A}, ID_F)$ where $sk_{ID_A} \leftarrow \text{Extract}^A(msk, ID_A)$.
- **Signing Query:** When $\mathcal{A}$ queries for a signature on a message m by an aircraft ID_F belonging to airline ID_A , $\mathcal{C}$ returns $\sigma \leftarrow \text{Sign}(sk_{ID_F}, m)$ where $sk_{ID_F} \leftarrow \text{Extract}^F(sk_{ID_A}, ID_F)$.

Output phase: $\mathcal{A}$ outputs a forged signature σ^* on a message m^* with respect to airline identity ID_A^* and aircraft identity ID_F^* . $\mathcal{A}$ wins the game if the following conditions hold:

- 1) $\text{Verify}(\sigma^*, ID_A^*, ID_F^*, m^*) = 1$;
- 2) $\mathcal{A}$ never made Extract^A Query on ID_A^* .
- 3) $\mathcal{A}$ never made Extract^F Query on (ID_A^*, ID_F^*) .
- 4) $\mathcal{A}$ never made Signing Query on (ID_A^*, ID_F^*, m^*) .

Definition 2. An ADS-B HIBS is said to be $(t, q_A, q_F, q_S, \epsilon)$ -existential unforgeable against selective ID chosen message attacks, if $\mathcal{A}$ cannot win the game above within time t and with probability greater than ϵ , by making q_A Extract^A Queries, q_F Extract^F Queries, and q_S Signing Queries, respectively.

Remark 1. Our defined security model is a general model for HIBS and it does not capture batch verification. Intuitively, a batch verification scheme requires (1) If all the n signatures $\sigma_1, \dots, \sigma_n$ pass the **Verify** algorithm, then they also pass the **BVerify** algorithm. (2) If any of the n signatures does not pass the **Verify** algorithm, then the n signatures will fail to pass the **BVerify** algorithm with overwhelming probability. Later we will see that our proposed schemes support batch verification, but how to prove the security of batch verification under a model for batch verification is an interesting open problem.

4 PROPOSED ADS-B HIBS SCHEMES

4.1 The Basic Scheme

We first propose a basic scheme that supports partial batch verification. Partial batch verification refers to that the signatures to be verified should be generated by aircraft belonging to the same airline. We apply the Small Exponents Test technology proposed by Bellare et al. [4] to our batch verification in both the basic and the extended schemes to guarantee the security of the batch verification, since otherwise it is possible to forge two signatures σ'_1 and σ'_2 from two valid signatures σ_1 and σ_2 such that σ'_1 and σ'_2 can pass the **BVerify** algorithm although individual signatures of them does not pass the **Verify** algorithm.

Below is our proposed basic ADS-B HIBS scheme:

- 1) **Setup**: Let $\mathbb{G}$ and $\mathbb{G}_T$ be two cyclic groups of order p where $|p| \geq \kappa$, $e: \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ be an admissible bilinear map, g be a generator of $\mathbb{G}$. Let $H_1, H_2: \{0, 1\}^* \rightarrow \mathbb{G}$, $H_3: \{0, 1\}^* \rightarrow \mathbb{Z}_p$ be three cryptographic hash functions. The root PKG selects at random $\alpha \in \mathbb{Z}_p$, sets $msk = \alpha$, computes $P_{pub} = g^\alpha$, and sets $mpk = (1^\kappa, p, \mathbb{G}, \mathbb{G}_T, e, g, H_1, H_2, H_3, P_{pub})$.
- 2) **Extract^A**: The root PKG sets $sk_{ID_A} = H_1(ID_A)^\alpha$, and sends it to the airline with identity ID_A through a private channel.
- 3) **Extract^F**: An airline with identity ID_A picks at random $s \in \mathbb{Z}_p$, sets $sk_{ID_F} = (H_2(ID_A || ID_F) \cdot sk_{ID_A})^s$, and computes $P = H_1(ID_A)^s$, $R = H_2(ID_A || ID_F)^s$, and sends (sk_{ID_F}, P, R) to the aircraft with identity ID_F through a private channel. Note that P and R are parts of the signing keys as well as the signatures. We need them in the verification procedure.
- 4) **Sign**: An aircraft with identity ID_F belonging to airline ID_A picks at random $\gamma \in \mathbb{Z}_p$, computes $U = g^\gamma$, $h = H_3(ID_A || ID_F || m || U || P || R)$, and $V = sk_{ID_F} \cdot P_{pub}^{h\gamma}$. The signature is $\sigma = (U, V, P, R)$ on message m .
- 5) **Verify**: A signature $\sigma = (U, V, P, R)$ on message m with respect to airline identity ID_A and aircraft identity ID_F can be verified as follows:
 - 1) Check if

$$e(P, H_2(ID_A || ID_F)) \stackrel{?}{=} e(H_1(ID_A), R);$$

- 2) Check if

$$e(V, g) \stackrel{?}{=} e(R, g) \cdot e(P \cdot U^h, P_{pub}),$$

$$\text{where } h = H_3(ID_A || ID_F || m || U || P || R).$$

If both the checkings hold, output 1, otherwise output 0.

- 6) **BVerify**: A list of signatures $\{\sigma_i = (U_i, V_i, P_i, R_i)\}_{i=1}^n$ on messages $\{m_i\}_{i=1}^n$ with respect to the same airline identity ID_A (the special case of $\{ID_{A_i}\}_{i=1}^n$ where $ID_{A_i} = ID_{A_j}$ for any $i, j \in [1, n]$) and aircraft identities $\{ID_{F_i}\}_{i=1}^n$ can be verified as follows:
 - 1) Check if $P_i \stackrel{?}{=} P_j$ for any $i, j \in [1, n]$;

- 2) Check if

$$e\left(P_1, \prod_{i=1}^n H_2(ID_A || ID_{F_i})\right) \stackrel{?}{=} e(H_1(ID_A), \prod_{i=1}^n R_i);$$

- 3) Select n random numbers of ℓ_b bits $\delta_1, \dots, \delta_n$, where ℓ_b is a small number (e.g., 80). Check if

$$e\left(\prod_{i=1}^n V_i^{\delta_i}, g\right) \stackrel{?}{=} e\left(\prod_{i=1}^n P_i^{\delta_i} \cdot U_i^{h_i \delta_i}, P_{pub}\right) \cdot e\left(\prod_{i=1}^n R_i^{\delta_i}, g\right),$$

$$\text{where } h_i = H_3(ID_A || ID_{F_i} || m_i || U_i || P_i || R_i).$$

If all the checking hold, output 1, otherwise output 0.

Correctness of the proposed ADS-B HIBS with partial batch verification:

$$\begin{aligned} e\left(\prod_{i=1}^n V_i^{\delta_i}, g\right) &= e\left(\prod_{i=1}^n (sk_{ID_{F_i}} \cdot P_{pub}^{h_i \gamma_i})^{\delta_i}, g\right) \\ &= e\left(\prod_{i=1}^n (H_2(ID_A || ID_{F_i}) \cdot sk_{ID_A})^{s \delta_i} \cdot P_{pub}^{h_i \gamma_i \delta_i}, g\right) \\ &= e\left(\prod_{i=1}^n R_i^{\delta_i}, g\right) \cdot e\left(\prod_{i=1}^n H_1(ID_A)^{s \delta_i} \cdot g^{h_i \gamma_i \delta_i}, g^\alpha\right) \\ &= e\left(\prod_{i=1}^n R_i^{\delta_i}, g\right) \cdot e\left(\prod_{i=1}^n P_i^{\delta_i} \cdot U_i^{h_i \delta_i}, P_{pub}\right) \\ &= e\left(\prod_{i=1}^n R_i^{\delta_i}, g\right) \cdot e\left(\prod_{i=1}^n P_i^{\delta_i} \cdot U_i^{h_i \delta_i}, P_{pub}\right). \end{aligned}$$

4.2 Extension for Full Batch Verification

Partial batch verification might be not sufficient in practice as the signatures are generally produced by aircraft belonging to different airlines rather than the same airline. To get over this barrier, a useful observation is that if we assume P_i and R_i of the signatures $\{\sigma_i = (U_i, V_i, P_i, R_i)\}_{i=1, \dots, n}$ with respect to airline identities $\{ID_{A_i}\}_{i=1}^n$ and aircraft identities $\{ID_{F_i}\}_{i=1}^n$ are always valid, in particular, if:

$$e(P_i, H_2(ID_{A_i} || ID_{F_i})) = e(H_1(ID_{A_i}), R_i) \quad (1)$$

always hold for $i = 1, \dots, n$, then the n signatures can be verified as below:

$$e\left(\prod_{i=1}^n V_i, g\right) \stackrel{?}{=} e\left(\prod_{i=1}^n R_i, g\right) \cdot e\left(\prod_{i=1}^n P_i \cdot U_i^{h_i}, P_{pub}\right).$$

We assume that some certificate authority is available to assure the correspondence between the airline/aircraft's identities and the (P_i, R_i) . In particular, it is assumed that $P = sk_{ID_{F,2}}$ and $R = sk_{ID_{F,3}}$ in sk_{ID_F} are certified by a CA such that the checking (1) above is always guaranteed. Then it is obvious to see that we can support full batch verification regardless of the airlines as in the following extended scheme of ADS-B HIBS:

- 1) **Setup:** Same as in the basic scheme.
- 2) **Extract^A:** Same as in the basic scheme.
- 3) **Extract^F:** An airline with identity ID_A picks at random $s \in \mathbb{Z}_p$, sets $sk_{ID_F} = (H_2(ID_A \| ID_F) \cdot sk_{ID_A})^s$ and sends it to the aircraft with identity ID_F through a private channel. Then the airline ID_A publishes $P = H_1(ID_A)^s$ and $R = H_2(ID_A \| ID_F)^s$ as certification of ID_F belonging to ID_A .
- 4) **Sign:** An aircraft with identity ID_F belonging to airline ID_A picks at random $\gamma \in \mathbb{Z}_p$, computes $U = g^\gamma$, $h = H_3(ID_A \| ID_F \| m \| U)$, then computes $V = sk_{ID_F} \cdot P_{pub}^{h\gamma}$. The signature is $\sigma = (U, V)$ on message m .
- 5) **Verify:** A signature $\sigma = (U, V)$ on message m with respect to airline identity ID_A and aircraft identity ID_F can be verified as $e(V, g) \stackrel{?}{=} e(R, g) \cdot e(P \cdot U^h, P_{pub})$, where $h = H_3(ID_A \| ID_F \| m \| U)$, P and R corresponding to (ID_F, ID_A) are fetched from the published certification list.
- 6) **BVerify:** A list of signatures $\{\sigma_i = (U_i, V_i)\}_{i=1}^n$ on messages $\{m_i\}_{i=1}^n$ with respect to airline identities $\{ID_{A_i}\}_{i=1}^n$ and aircraft identities $\{ID_{F_i}\}_{i=1}^n$ can be verified as:
 - 1) Select n random numbers of ℓ_b bits $\delta_1, \dots, \delta_n$, where ℓ_b is a small number (e.g., 80).
 - 2) Compute $\hat{P} = \prod_{i=1}^n P_i^{\delta_i}$, $\hat{R} = \prod_{i=1}^n R_i^{\delta_i}$ where P_i 's and R_i 's corresponding to (ID_{F_i}, ID_{A_i}) are fetched from the published certification list.
 - 3) Check if $e(\prod_{i=1}^n V_i^{\delta_i}, g) \stackrel{?}{=} e(\hat{R}, g) \cdot e(\hat{P} \cdot \prod_{i=1}^n U_i^{h_i \delta_i}, P_{pub})$, where $h_i = H_3(ID_{A_i} \| ID_{F_i} \| m_i \| U_i)$.

The correctness of this extended version of ADS-B HIBS with full batch verification is similar to that of the basic scheme, so omitted here.

4.3 Discussions

4.3.1 Assumption on P and R

We discuss why we have to assume that P and R are certified by some existing techniques such as CA in the following.

The trick is that if we do not assume to use some existing techniques to verify the value of P and R , then we have to check the equation (1) before verifying the signatures. Namely, when multiple signatures are received, the following equation should be checked first:

$$\prod_{i=1}^n e(P_i, H_2(ID_{A_i} \| ID_{F_i})) = \prod_{i=1}^n e(H_1(ID_{A_i}), R_i). \quad (2)$$

Unfortunately, equation (2) does not reduce the computation cost compared with checking n signatures independently. This violates our purpose that is to significantly improve the verification efficiency.

On the other hand, if we do not verify the value of P and R , then the adversary could forge any valid signatures as following.

The adversary first chooses at random $t, r \in \mathbb{Z}_p$, and sets $P = g^t$, $U = g^r$. Then select at random $R \in \mathbb{G}$. Finally compute $h = H_3(ID_A \| ID_F \| m \| U)$ and $V = R \cdot P_{pub}^{t+rh}$. It is easy to verify that (U, V, P, R) can pass the verification algorithm.

We leave it as a future work to remove the assumption and

at the same time the verification of P and R can be done efficiently.

4.3.2 Compromise of CA's Key

In the extended scheme we need PKI or similar for certifying some parameters such as P and R . Thus this scheme shares the same problem of key compromise with PKI systems. If the private keys of the CA are compromised, all certificates for P and R should be recalled and then reissued with a new CA key.

4.3.3 Communication Overhead

The communication overheads are $4|G|$ and $2|G|$ in the basic and extended scheme, respectively, where $|G|$ is the bit length of an element in group G . For example, if $|G|=512$, then the communication costs will be 2,048 bits and 1,024 bits respectively, both of which are much larger than the maximum payloads of ADS-B Out message specified in 1090ES (56 bits) or UAT (256 bits). According to [24], we can either send the signatures separately (one ADS-B standard message followed with several ADS-B messages that package the divided signature segments in the ADS-B data blocks), or send one message which includes an ADS-B message and its signature, where the latter method requires that the ADS-B message format could be altered.

4.3.4 Identification of Invalid Signatures

An ADS-B receiver (whether it is the aircraft or ground one) could receive a large number of ADS-B messages together with their signatures, which are sent by the surrounding aircraft every second. In case that there is an invalid signature in this short period time (i.e., within second), it may not be so imperative to find out which signatures are incorrect since the new valid signatures on new messages are coming in another second (Note that these messages contain the information of the aircraft such as position, altitude, and speed, and there will not be much change on these values from previous one sent half a second ago). Nevertheless, if there are too many invalid signatures, it may indicate that the signatures are forged by malicious adversaries. If we want to find out which signatures are forged ones, we can use a recursive divide-and-conquer approach. In particular, if the batch verification fails, we can divide these signatures into two halves and repeat the batch verification on them. If the first half passes the batch verification algorithm, it means the invalid signatures are in the second half. We can repeat this step until we find out the invalid signatures.

4.3.5 Replay Attack

The eavesdroppers can intercept and replay the messages as well as the corresponding signatures. However these attackers cannot forge new signatures even they have eavesdropped valid signatures. Theoretically, it is trivial to prevent the replay attack: we can simply append a timestamp to the message, in which case the replayed signatures are authentic but outdated. While it is trivial from a theoretical point of view, it is not from a real-world system perspective in that adding timestamps will increase the memory overhead. Nevertheless, adding timestamps is a

really important consideration for future air traffic communication protocol designing.

5 SECURITY PROOF

5.1 Complexity Assumption

The proposed ADS-B HIBS is based on the randomized Computational Diffie-Hellman assumption (RCDH) as defined below. This problem is conjectured to be as hard as CDH problem.

Definition 3 (RCDH Assumption). Let $\mathbb{G}$ and $\mathbb{G}_T$ be two cyclic groups of order p where $|p| \geq \kappa$, and $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$ is an admissible bilinear map, g is a generator of $\mathbb{G}$. We say that (ϵ, t) -RCDH assumption holds in $\mathbb{G}$ if for any PPT adversary $\mathcal{A}$ running within time t , there exists a negligible function ϵ such that for any security parameter $\kappa \in \mathbb{N}$:

$$\Pr[(Z, Z^{xy}) \in \mathbb{G}^2 \leftarrow \mathcal{A}(1^\kappa, p, \mathbb{G}, \mathbb{G}_T, e, g, X, Y) : X = g^x, Y = g^y] \leq \epsilon(\kappa),$$

where x, y are chosen at random from $\mathbb{Z}_p$ and Z are selected by $\mathcal{A}$ at random from $\mathbb{G}$.

To gain confidence in the hardness of this new assumption, we can show that it achieves desirable intractability in the generic group model.

In the generic group model, elements of $\mathbb{G}$ and $\mathbb{G}_T$ are encoded as unique random strings in $\{0, 1\}^*$, so that no property other than equality can be directly tested by the adversary. For solving the instance of RCDH problem, three oracles are provided to perform the group operation between group elements in $\mathbb{G}$ (resp. $\mathbb{G}_T$) as well as the bilinear pairing $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$. The opaque encoding of the elements of $\mathbb{G}$ is modeled as an injective function $\xi : \mathbb{Z}_p \rightarrow \mathcal{E} \subset \{0, 1\}^*$, where $|\mathcal{E}| = p$. It allows mapping of all $a \in \mathbb{Z}_p$ to $\xi(a) = g^a \in \mathbb{G}$. Similarly we can define $\xi_T : \mathbb{Z}_p \rightarrow \mathcal{E}_T$ for $\mathbb{G}_T$. The attacker $\mathcal{A}$ communicates with the oracles using the representations of the group elements only. Then we have the Theorem 1 as following.

Theorem 1. Let $\mathcal{A}$ be an adversary that breaks RCDH assumption in the generic group model, making totally q_G queries to the oracles computing the group operations in $\mathbb{G}$ and $\mathbb{G}_T$ as well as the pairing operation $e : \mathbb{G} \times \mathbb{G} \rightarrow \mathbb{G}_T$. If $x, y \in \mathbb{Z}_p$ and ξ, ξ_T are chosen at random, then the probability that $(\xi(s), \xi_T(xy)) \leftarrow \mathcal{A}(p, \xi(1), \xi(x), \xi_T(y))$ where $s \in \mathbb{Z}_p$, is bounded by:

$$\epsilon \leq \frac{(q_G + 3)^2}{p} = O\left(\frac{(q_G)^2}{p}\right).$$

Proof. In order to answer oracle queries made by $\mathcal{A}$, we initiate two lists of pairs $L_G = \{(F_i, \xi_i)\}$ for $\mathbb{G}$ and $L_{G_T} = \{(F_{T,i}, \xi_{T,i})\}$ for $\mathbb{G}_T$, and set $F_0 = 1, F_1 = x, F_2 = y$ while ξ_0, ξ_1, ξ_2 are set to distinct strings in $\mathcal{E}$ at random, and add $(F_0, \xi_0), (F_1, \xi_1), (F_2, \xi_2)$ into L_G . Then $\mathcal{A}$ can queries the oracles below:

- **GroupOperation Query:** When $\mathcal{A}$ queries for multiplication or division of two operands ξ_i, ξ_j in the list L_G where $|L_G| = \tau$ (initially $\tau = 3$), we set $\tau = \tau + 1$, compute $F_\tau = F_i \pm F_j \in \mathbb{Z}_p$ depending

on whether the requested operation is multiplication or division in $\mathbb{G}$. If $F_\tau = F_l$ for some $l < \tau$, we set $\xi_\tau = \xi_l$, otherwise we set ξ_τ as an arbitrary string in $\mathcal{E}$ that is distinct from $\xi_0, \dots, \xi_{\tau-1}$. Then we add (F_τ, ξ_τ) to the list L_G and returns ξ_τ .

Group operation in $\mathbb{G}_T$ is treated similarly using L_{G_T} .

- **Pairing Query:** when $\mathcal{A}$ queries for pairing of two operands ξ_i, ξ_j in the list L_G , we set $\tau_T = \tau_T + 1$ where $\tau_T = |L_{G_T}|$ (initially $\tau_T = 0$), compute $F_{T,\tau_T} = F_i \cdot F_j \in \mathbb{Z}_p$. If $F_{T,\tau_T} = F_{T,l}$ for some $l < \tau_T$, we set $\xi_{T,\tau_T} = \xi_{T,l}$, otherwise we set ξ_{T,τ_T} as an arbitrary string in $\mathcal{E}_T$ that is distinct from $\xi_{T,0}, \dots, \xi_{T,\tau_T-1}$. Then we add $(F_{T,\tau_T}, \xi_{T,\tau_T})$ to the list L_{G_T} and returns ξ_{T,τ_T} .

The output of $\mathcal{A}$ is a pair (ξ_{l_1}, ξ_{l_2}) when it terminates, where $l_1, l_2 \in [0, \tau]$. If this pair implies a valid solution of the given RCDH instance, then we have $F_{l_2} = xyF_{l_1}$ where $(F_{l_1}, \xi_{l_1}), (F_{l_2}, \xi_{l_2}) \in L_G$. In particular,

$$F_{l_2}(x, y) - xyF_{l_1}(x, y) = 0,$$

where F_{l_2} and F_{l_1} are polynomials on x, y of degree at most 1. As xyF_{l_1} is a polynomial of degree at least 2, we can deduce that $F_{l_2} - xyF_{l_1}$ is a non-trivial polynomial that admits at most three roots. Let x^*, y^* are two random elements in $\mathbb{Z}_p$, then the success probability of $\mathcal{A}$ is bounded by the probability of the union of the following events:

- 1) $F_i(x^*, y^*) - F_j(x^*, y^*) = 0$ for some $i, j \in [0, \tau]$ such that $F_i \neq F_j$;
- 2) $F_{T,i}(x^*, y^*) - F_{T,j}(x^*, y^*) = 0$ for some $i, j \in [0, \tau_T]$ such that $F_{T,i} \neq F_{T,j}$;
- 3) $F_{l_2}(x^*, y^*) - x^*y^*F_{l_1}(x^*, y^*) = 0$.

Since $F_i - F_j$ for fixed i, j is a polynomial of degree at most 1, it vanishes at a random $x^*, y^* \in \mathbb{Z}_p$ with probability at most $1/p$. Similarly, $F_{T,i} - F_{T,j}$ for fixed i, j is a polynomial of degree at most 2, it vanishes at a random $x^*, y^* \in \mathbb{Z}_p$ with probability at most $2/p$. And for the third case $F_{l_2} - xyF_{l_1}$, it is a polynomial of degree at most 3 that vanishes with probability at most $3/p$. Conclusively, we can see that $\mathcal{A}$ solves the RCDH problem with advantage $\epsilon \leq C(\tau, 2) * 1/p + C(\tau_T, 2) * 2/p + 3/p$. As $\tau + \tau_T \leq q_G + 3$, we have the upper bound of advantage as $\epsilon \leq (q_G + 3)^2/p = O((q_G)^2/p)$. $\square$

5.2 Security Proof of the Extended Scheme

In this section, we give a formal security proof for the proposed extended scheme that supports full batch verification. Note that the security proof for the basic scheme is similar to that for the extended scheme and thus we omit it for saving space.

The proof employs the *reduction* technique, in which we will show how to convert any efficient adversary $\mathcal{A}$ that succeeds in breaking the ADS-B scheme with non-negligible probability into an efficient algorithm $\mathcal{B}$ that can succeed in solving the RCDH problem that has been proved hard. Intuitively, $\mathcal{B}$ plays the game with $\mathcal{A}$ by simulating the challenger $\mathcal{C}$ with its own RCDH instance, aiming to solve the RCDH problem.

Theorem 2. The proposed ADS-B HIBS in Section 4.2 is $(t, q_A, q_F, q_S, \epsilon)$ -existential unforgeable against selective ID chosen message attacks in the random oracle model, if (ϵ', t') -RCDH assumption holds in $\mathbb{G}$:

$$\epsilon' \geq \epsilon(1 - q_{H_3}/p)$$

$$t' \leq 120686q_{H_3}t/\epsilon + (q_{H_1} + q_{H_2} + 2q_A + 4q_F + 4q_S)t_{sm},$$

where t_{sm} is the time consumed for one scalar multiplication on $\mathbb{G}$, q_{H_1} , q_{H_2} and q_{H_3} are the number of queries made to the random oracle H_1 , H_2 and H_3 respectively.

Proof. Suppose the ADS-B HIBS is broken by a PPT adversary $\mathcal{A}$, we can construct a PPT algorithm $\mathcal{B}$ that outputs the solution of RCDH instances in $\mathbb{G}$ with the advantage described in the theorem. Let the given RCDH instance be $(1^\kappa, p, \mathbb{G}, \mathbb{G}_T, e, g, X = g^x, Y = g^y)$, then $\mathcal{B}$'s goal is to compute $(Z, Z^{xy}) \in \mathbb{G}^2$. In particular, $\mathcal{B}$ plays the existential unforgeability game with $\mathcal{A}$, by simulating the behavior of the challenger $\mathcal{C}$ as below:

Setup Phase: After receiving ID_F^* and ID_A^* from $\mathcal{A}$, $\mathcal{B}$ initiates two empty lists L_{H_1} , L_{H_2} and performs the following computation:

- 1) Sets $P_{pub} = X$ (which implies that $\alpha = x$), and sets $H_{ID_A^*} = Y$.
- 2) Inserts the entry $(ID_A^*, H_{ID_A^*})$ into L_{H_1} .
- 3) Picks at random $\tau^* \in \mathbb{Z}_p$, and sets $H_{ID_A^*||ID_F^*} = g^{\tau^*}$.
- 4) Inserts $(ID_A^*||ID_F^*, H_{ID_A^*||ID_F^*}, \tau^*)$ into L_{H_2} .

Then $\mathcal{B}$ picks three hash functions $H_1, H_2 : \{0, 1\}^* \rightarrow \mathbb{G}$, $H_3 : \{0, 1\}^* \rightarrow \mathbb{Z}_p$ that are modeled as random oracles, and returns $mpk = (1^\kappa, p, \mathbb{G}, \mathbb{G}_T, e, g, H_1, H_2, H_3, P_{pub})$ to $\mathcal{A}$.

Query Phase: At the beginning of this phase, $\mathcal{B}$ additionally initiates an empty list L_{H_3} . Then $\mathcal{B}$ answers the following queries.

- **H₁ Query:** When $\mathcal{A}$ queries for the output of H_1 on an input M , $\mathcal{B}$ checks if there exists an entry (M, H_M, θ) in L_{H_1} . If yes, $\mathcal{B}$ returns H_M ; otherwise, $\mathcal{B}$ picks at random $\theta \in \mathbb{Z}_p$, sets $H_M = g^\theta$, and inserts (M, H_M, θ) into L_{H_1} and returns H_M .
- **H₂ Query:** When $\mathcal{A}$ queries for the output of H_2 on an input M , $\mathcal{B}$ checks if there exists an entry (M, H_M, τ) in L_{H_2} . If yes, $\mathcal{B}$ returns H_M ; otherwise, $\mathcal{B}$ picks at random $\tau \in \mathbb{Z}_p$, sets $H_M = Y^\tau$, and inserts (M, H_M, τ) into L_{H_2} and returns H_M .
- **H₃ Query:** When $\mathcal{A}$ queries for the output of H_3 on an input M , $\mathcal{B}$ checks if there exists an entry (M, H_M) in L_{H_3} . If yes, $\mathcal{B}$ returns H_M ; otherwise, $\mathcal{B}$ picks at random $H_M \in \mathbb{Z}_p$, inserts (M, H_M) in L_{H_3} and returns H_M .
- **Extract^A Query:** When $\mathcal{A}$ queries for the secret key of an identity $ID_A \neq ID_A^*$, $\mathcal{B}$ checks if there exists an entry (ID_A, H_{ID_A}, θ) in L_{H_1} ; if not, $\mathcal{B}$ picks at random $\theta \in \mathbb{Z}_p$, sets $H_{ID_A} = g^\theta$, and inserts (ID_A, H_{ID_A}, θ) into L_{H_1} . Then $\mathcal{B}$ returns $sk_{ID_A} = X^\theta$. This simulation is perfect, because: $sk_{ID_A} = X^\theta = g^{x\theta} = (g^\theta)^x = H_1(ID_A)^x = H_1(ID_A)^\alpha$

Extract^F Query: When $\mathcal{A}$ queries for the secret key of an aircraft identity ID_F belonging to an airline identity ID_A , $\mathcal{B}$ performs as follows.

- If $ID_A \neq ID_A^*$, then $\mathcal{B}$ generates the secret key sk_{ID_A} for ID_A as in the simulation of Extract^A Query above, and runs $sk_{ID_F} \leftarrow \text{Extract}^F(sk_{ID_A}, ID_F)$. Then $\mathcal{B}$ returns sk_{ID_F} , publishes (P, R) and records the corresponding $s \in \mathbb{Z}_p$ used in the algorithm Extract^F.
- If $ID_A = ID_A^*$ and $ID_F \neq ID_F^*$, then $\mathcal{B}$ performs as follows:
 - 1) Checks if there exists an entry $(ID_A^*||ID_F, H_{ID_A^*||ID_F}, \tau)$ in L_{H_2} ; if not, $\mathcal{B}$ picks at random $\tau \in \mathbb{Z}_p$, sets $H_{ID_A^*||ID_F} = Y^\tau$, and inserts $(ID_A^*||ID_F, H_{ID_A^*||ID_F}, \tau)$ into L_{H_2} .
 - 2) Picks at random $\xi \in \mathbb{Z}_p$, sets $P = g^\xi$, $R = P^\tau$, $sk_{ID_F} = R \cdot X^\xi$, returns sk_{ID_F} and publishes (P, R) .

This simulation is perfect, because denote $s = y^{-1} \xi \in \mathbb{Z}_p$, then we have:

$$P = g^\xi = g^{ys} = Y^s = H_1(ID_A^*)^s$$

$$R = P^\tau = (Y^s)^\tau = (Y^\tau)^s = H_2(ID_A^*||ID_F^*)^s$$

$$sk_{ID_F} = R \cdot X^\xi = R \cdot P_{pub}^\xi = R \cdot g^{xys}$$

$$= R \cdot (Y^x)^s = R \cdot (H_1(ID_A^*)^x)^s$$

$$= R \cdot (sk_{ID_A^*})^s$$

$$= (H_2(ID_A^*||ID_F) \cdot sk_{ID_A^*})^s$$

Hence the returned secret key sk_{ID_F} for ID_F is always valid.

Signing query: When $\mathcal{A}$ queries for a signature with respect to a pair of aircraft-airline identities (ID_F, ID_A) on a message m , $\mathcal{B}$ performs the following computations:

- 1) If $ID_A \neq ID_A^*$ or $ID_F \neq ID_F^*$, $\mathcal{B}$ generates the secret key sk_{ID_F} for ID_F as in the simulation of Extract^F Query above, and runs $\sigma \leftarrow \text{Sign}(sk_{ID_F}, m)$ and returns σ .
- 2) If $ID_A = ID_A^*$ and $ID_F = ID_F^*$, $\mathcal{B}$ picks at random $s \in \mathbb{Z}_p$, computes $P = H_1(ID_A^*)^s$, $R = H_2(ID_A^*||ID_F^*)^s$, then picks $h \in \mathbb{Z}_p$, computes $U = g/P^{h-1}$, $V = R \cdot P_{pub}^h$, inserts $(ID_A^*||ID_F^*||m||U, h)$ in L_{H_3} and returns $\sigma = (U, V)$ and publishes (P, R) . The simulation is perfect, because denote $\gamma = 1 - ysh^{-1}$, we have:

$$U = \frac{g}{P^{h-1}} = \frac{g}{(Y^s)^{h-1}} = \frac{g}{g^{ys(h-1)}}$$

$$= g^{1-ysh^{-1}} = g^\gamma$$

$$V = R \cdot P_{pub}^h = R \cdot P_{pub}^{ys+h(1-ysh^{-1})}$$

$$= R \cdot P_{pub}^{ys} \cdot P_{pub}^{h\gamma} = R \cdot g^{xys} \cdot P_{pub}^{h\gamma}$$

$$= R \cdot (Y^x)^s \cdot P_{pub}^{h\gamma} = R \cdot (sk_{ID_A^*})^s \cdot P_{pub}^{h\gamma}$$

$$= (H_2(ID_A^*||ID_F^*) \cdot sk_{ID_A^*})^s \cdot P_{pub}^{h\gamma}$$

$$= sk_{ID_F^*} \cdot P_{pub}^{h\gamma}$$

It is stressed that if there already exists $(ID_A^* \| ID_F^* \| m \| U, h')$ in L_{H_3} where $h' \neq h$, $\mathcal{B}$ aborts the simulation. However, this happens with negligible probability q_{H_3}/p , because U is uniformly distributed in $\mathbb{G}$ due to the randomness of $h \in \mathbb{Z}_p$.

Output phase: $\mathcal{A}$ outputs a forgery $\sigma^* = (U^*, V^*)$ on messages m^* with respect to airline identity ID_A^* and aircraft identity ID_F^* .

Suppose (P^*, R^*) was the published certificate corresponding to (ID_A^*, ID_F^*) , we have:

$$e(P^*, H_2(ID_A^* \| ID_F^*)) = e(R^*, H_1(ID_A^*))$$

Or equivalently,

$$e(P^*, g^{t^*}) = e(R^*, Y)$$

where $P^* = H_1(ID_A^*)^s$ and $R^* = H_2(ID_A^* \| ID_F^*)^s$.

Since $e(P^*, g^{t^*}) = e((P^*)^{t^*}, g)$ and $e(R^*, Y) = e(R^*, g^y) = e((R^*)^y, g)$, we have $(R^*)^{(t^*)^{-1}} = (P^*)^{y^{-1}}$. Then $\mathcal{B}$ computes $Z = (R^*)^{(t^*)^{-1}} \in \mathbb{G}$, and the next step is to compute $Z^{xy} = (P^*)^{y^{-1}xy} = (P^*)^x$.

Denote $h^* = H_3(ID_A^* \| ID_F^* \| m^* \| U^*)$, then we have:

$$\begin{aligned} V^* &= (H_2(ID_A^* \| ID_F^*) \cdot sk_{ID_A^*})^s \cdot P_{pub}^{h^* \cdot \gamma^*} \\ &= H_2(ID_A^* \| ID_F^*)^s \cdot (sk_{ID_A^*})^s \cdot g^{x h^* \cdot \gamma^*} \\ &= H_2(ID_A^* \| ID_F^*)^s \cdot (H_1(ID_A^*)^x)^s \cdot g^{x h^* \cdot \gamma^*} \\ &= R^* \cdot (P^* \cdot (U^*)^{h^*})^x \end{aligned}$$

$\mathcal{B}$ rewinds $\mathcal{A}$ to the time point when $\mathcal{A}$ queried $ID_A^* \| ID_F^* \| m^* \| U^*$ to the random oracle H_3 , and returns another value $h'^* \neq h^*$. By the forking lemma, $\mathcal{A}$ will return another valid forgery $\sigma' = (U^*, V'^*)$ within expected time $\tilde{t} = 120686q_{H_3}t/\epsilon$ where $\epsilon \geq 10(q_S + q_{H_3})(q_S + 1)/p$, such that:

$$V'^* = R^* \cdot (P^* \cdot (U^*)^{h'^*})^x.$$

Thus we have:

$$\begin{aligned} \frac{(V^*)^{(h^*)^{-1}}}{(V'^*)^{(h'^*)^{-1}}} &= \frac{(R^*)^{(h^*)^{-1}} \cdot (P^*)^{x(h^*)^{-1}} \cdot (U^*)^x}{(R^*)^{(h'^*)^{-1}} \cdot (P^*)^{x(h'^*)^{-1}} \cdot (U^*)^x} \\ &= (R^*)^{(h^*)^{-1} - (h'^*)^{-1}} \cdot ((P^*)^x)^{(h^*)^{-1} - (h'^*)^{-1}}. \end{aligned}$$

Then $\mathcal{B}$ can compute Z^{xy} as:

$$\begin{aligned} Z^{xy} &= (P^*)^x \\ &= \left(\frac{(V^*)^{(h^*)^{-1}}}{(V'^*)^{(h'^*)^{-1}} \cdot (R^*)^{(h^*)^{-1} - (h'^*)^{-1}}} \right)^{((h^*)^{-1} - (h'^*)^{-1})^{-1}}. \end{aligned}$$

Finally, $\mathcal{B}$ outputs (Z, Z^{xy}) as the solution of the given RCDH instance. As we can see, $\mathcal{B}$ can complete the simulation without abort by the advantage of $\epsilon' \geq \epsilon(1 - q_{H_3}/p)$, and the running time is about $t' = \tilde{t} + (q_{H_1} + q_{H_2} + 2q_A + 4q_F + 4q_S)t_{sm} = 120686q_{H_3}t/\epsilon + (q_{H_1} + q_{H_2} + 2q_A + 4q_F + 4q_S)t_{sm}$. That is, if $\mathcal{A}$ can break the ADS-B HIBS experiment with the advantage at least ϵ , then the

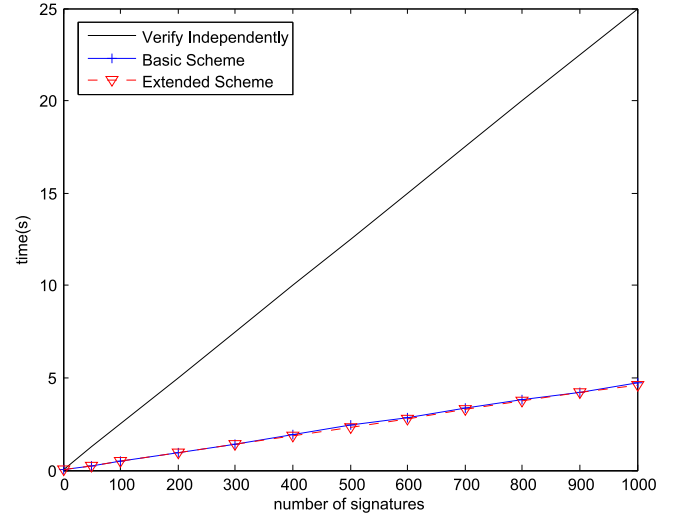


Fig. 3. The comparison of running time of verifying n signatures.

advantage that $\mathcal{B}$ can solve the RCDH problem is at least ϵ' within the running time of t' . This completes the proof. $\square$

6 PERFORMANCE

We implement our proposed schemes on a desktop PC with a 3.16 GHz Intel Core(TM)2 Duo CPU and 4GB memory. We use the pairing-based cryptography (PBC) library basing on C language. We choose the Type-A pairing, with the base field size of 512-bits, and is constructed on the curve $y^2 = x^3 + x$, the degree of which is $k = 2$ and supports Discrete-Logarithm security of 1,024-bits. Type-A pairing is symmetric and thus is suitable for our schemes.

In the experiment, we run each scheme ten times to get an average value. We measure the running time of generating and verifying n signatures where n could be from 1 to 1,000 increment by 100. We test for three schemes, that is: the scheme verifying n signatures independently which means verifying individual signatures in the proposed basic scheme, the basic scheme (supporting partial batch verification) and the extended scheme (support full batch verification). The result shows that generating a signature requires about 15 milliseconds (ms). The verification time is shown in Fig. 3. From the experiment results, we can see that when $n > 10$, both our basic scheme and extended scheme are about five times faster than verifying n signatures independently. This is because that in both the basic scheme and the extended scheme that support batch verification, the number of the most time-consuming operations, i.e., pairings, keep to be a constant despite of n . The main difference between the basic scheme and the extended scheme is that in the basic scheme we need to check the validity of P and R which results in requiring two more pairings calculation, while in the extended scheme we assume P and R have already been verified by some existing techniques such as CA. Note that no matter how many signatures are verified in batch at one time, we always need only two more pairings in the basic scheme than that in the full scheme. There are also some more multiplication operations needed in the basic scheme, but these operations are much less

time-consuming than the pairing operations and thus have little influence on the computational time. In particular, the time needed to compute a pairing is roughly 1,000 times longer than that of computing a multiplication. Therefore, as shown in Fig. 3, it appears that the basic and full schemes have similar computational time.

7 CONCLUSION

In this paper, we proposed a new ADS-B authentication framework based on three-level hierarchical identity-based signature with batch verification, which can significantly reduce the verification cost. Basing on the framework, we demonstrated two concrete schemes. The basic scheme supports only partial batch verification while the extended scheme provides full batch verification but requiring existing techniques to ensure the integrity of some public values. We gave a formal proof for the security of our extended scheme. The experiment results show that our proposed scheme is much more efficient than traditional ones.

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A. Yang is the corresponding author.

REFERENCES

- [1] [Online]. Available: <http://www.flightradar24.com>, 2015.
- [2] J. Baek, Y. J. Byon, E. Hableel, and M. Al-Qutayri, "An authentication framework for automatic dependent surveillance-broadcast based on online/offline Identity-based signature," in *Proc. 8th Int. Conf. P2P, Parallel, Grid, Cloud Internet Comput.*, 2013, pp. 358–363.
- [3] J. Baek, Y. J. Byon, E. Hableel, and M. Al-Qutayri, "Making air traffic surveillance more reliable: A new authentication framework for automatic dependent surveillance-broadcast (ADS-B) based on online/offline Identity-based signature," *Security Commun. Netw.*, vol. 8, no. 5, pp. 740–750, Mar. 2015.
- [4] M. Bellare, J. A. Garay, and T. Rabin, "Fast batch verification for modular exponentiation and digital signatures," in *Proc. Int. Conf. Theory Appl. Cryptographic Techn. Adv. Cryptol.*, 1998, vol. 1403, pp. 236–250.
- [5] D. Boneh, B. Lynn, and H. Shacham, "Short signatures from the weil pairing," *J. Cryptol.*, vol. 17, no. 4, pp. 297–319, 2004.
- [6] J. Camenisch, S. Hohenberger, and M. Ø. Pedersen, "Batch verification of short signatures," in *Proc. 26th Annu. Int. Conf. Theory Appl. Cryptographic Techn.*, 2007, vol. 4515, pp. 246–263.
- [7] A. Costin and A. Francillon, "Ghost in the air (traffic): On insecurity of ADS-B protocol and practical attacks on ADS-B devices," *Black Hat USA*, 2012, <http://media.blackhat.com/bh-us-12/Briefings/Costin/BHUS12CostinGhostsInAirWP.pdf>
- [8] Z. Feng, W. Pan, and Y. Wang, "A data authentication solution of ADS-B system based on x.509 certificate," in *Proc. 27th Int. Congr. Aeronaut. Sci.*, 2010, p. ICAS2010-10.7.3.
- [9] A. L. Ferrara, M. Green, S. Hohenberger, and M. Ø. Pedersen, "Practical short signature batch verification," in *Proc. Cryptographers' Track RSA Conf. Topics Cryptol.*, 2009, vol. 5473, pp. 309–324.
- [10] A. Fiat, "Batch RSA," in *Proc. Adv. Cryptol.*, 1989, vol. 435, pp. 175–185.
- [11] C. Gentry and A. Silverberg, "Hierarchical Id-based cryptography," in *Proc. 8th Int. Conf. Theory Appl. Cryptol. Inf. Security: Adv. Cryptol.*, 2002, vol. 2501, pp. 548–566.
- [12] J. Krozel, D. Andrisani, M. A. Ayoubi, T. Hoshizaki, and C. Schwalm, "Aircraft ADS-B data integrity check," in *Proc. 4th Aviation Technol., Integr. Oper. Forum*, 2004, pp. 1–11.
- [13] M.-I. Gaviria, M. Leonardi, G. Galati, and J. B.-Tejedor, "Localization algorithms for multilateration (MLAT) systems in airport surface surveillance," *Signal, Image Video Process.*, vol. 1, pp. 1–10, Jan. 2014.
- [14] F. Mattern and C. Floerkemeier, "From the internet of computers to the internet of things," in *From Active Data Management to Event-Based Systems and More*. New York, NY, USA: Springer-Verlag, 2010, pp. 242–259.
- [15] D. McCallie, J. Butts, and R. Mills, "Security analysis of the ADS-B implementation in the next generation air transportation system," *Int. J. Critical Infrastruct. Protection*, vol. 4, no. 2, pp. 78–87, 2011.
- [16] L. Purton, H. Abbass, and S. Alam, "Identification of ADS-B system vulnerabilities and threats," in *Proc. Australian Transport Res. Forum*, 2010, pp. 1–16.
- [17] K. Sampigethaya and R. Poovendran, "Visualization & assessment of ADS-B security for green ATM," in *Proc. 29th Digital Avionics Syst. Conf.*, 2010, pp. 3.A.3-1–3.A.3-16.
- [18] K. Sampigethaya, R. Poovendran, and L. Bushnell, "A framework for securing future E-enabled aircraft navigation and surveillance," in *Proc. AIAA Infotech@Aerospace Conf.*, 2009, pp. 1–10.
- [19] K. Samuelson, E. Valovage, and D. Hall, "Enhanced ADS-B research," in *Proc. IEEE Aerospace Conf.*, 2006, pp. 1–7.
- [20] M. Schäfer, V. Lenders, and I. Martinovic, "Experimental analysis of attacks on next generation air traffic communication," in *Proc. 11th Int. Conf. Appl. Cryptography Netw. Security*, 2013, pp. 253–271.
- [21] M. Strohmeier, V. Lenders, and I. Martinovic, "Security of ADS-B: State of the art and beyond," CoRR abs/1307.3664, 2013, <http://arxiv.org/abs/1307.3664>
- [22] M. Strohmeier, V. Lenders, and I. Martinovic, "Lightweight location verification in air traffic surveillance networks," in *Proc. 1st ACM Workshop Cyber-Phys. Syst. Security*, 2015, pp. 49–60.
- [23] M. Strohmeier, M. Schäfer, V. Lenders, and I. Martinovic, "Realities and challenges of nextgen air traffic management: The case of ADS-B," *IEEE Commun. Mag.*, vol. 52, no. 5, pp. 111–118, May 2014.
- [24] K. D. Wesson, T. E. Humphreys, and B. L. Evans, "Can cryptography secure next generation air traffic surveillance?" *IEEE Security Privacy Mag.*, <http://users.ece.utexas.edu/~bevans/papers/2015/nextgen/index.html>, 2014.
- [25] H. Yoon, J. H. Cheon, and Y. Kim, "Batch verifications with Id-based signatures," in *Proc. 7th Int. Conf. Inf. Security Cryptol.*, 2004, pp. 233–248.



Anjia Yang received the BS degree from Jilin University in 2011. He is currently working toward the PhD degree in the Department of Computer Science, City University of Hong Kong. His research interests include RFID security and privacy, applied cryptography, and cloud computing.



Xiao Tan received the BS and MS degrees from Fudan University, in 2007 and 2010, respectively, and the PhD degree from the City University of Hong Kong, Hong Kong, in 2014. He is currently a lecturer in Hangzhou Normal University, China. His main research interests include cryptography and information security, in particular, digital signatures, and encryption schemes.



Joonsang Baek received the PhD degree from Monash University, Australia. He is an assistant professor in the Department of Electrical and Computer Engineering, Khalifa University of Science, Technology and Research (KUSTAR), UAE. Before joining KUSTAR, he was a scientist at the Institute for Infocomm Research (I2R), Singapore. His research areas are in the field of cryptography and information security. He has published his work in a number of reputable journals and conference proceedings. He has also served as chairs and program committee members for a number of international conferences on information security and cryptography.



Duncan S. Wong received the BEng degree from the University of Hong Kong in 1994, the MPhil degree from the Chinese University of Hong Kong in 1998, and the PhD degree from Northeastern University, Boston, MA, in 2002. He is currently an associate professor in the Department of Computer Science at the City University of Hong Kong and the director of the Security and Data Sciences of ASTRI. His primary research interest is cryptography; in particular, cryptographic protocols, encryption and signature schemes, and anonymous systems. He is also interested in other topics in information security, such as network wireless security database security, and security in cloud computing.